Episodic vs. Continuing Tasks

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User

what is difference between Episodic vs. Continuing task in MDP

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Assistant gpt-4o-mini

In the context of Markov Decision Processes (MDPs), the concepts of episodic and continuing tasks refer to the structure of the decision-making problem, particularly in terms of how episodes are defined and how rewards are received over time. Here's a breakdown of the differences between the two:

1. Episodic Tasks:

Definition: In episodic tasks, the decision-making process is divided into episodes, where each episode consists of a sequence of states, actions, and rewards, culminating in a terminal state.
Structure: Each episode starts with an initial state and ends after reaching a terminal state. After each episode finishes, the process can be reset to the initial state.
Reward Structure: Rewards are typically received only during a single episode. The goal is usually to maximize the cumulative rewards (which can often be summed simply) for each episode.
Example: A robot learning to navigate through a maze, where it has to find an exit (terminal state) within each trial. Once it finds the exit, that episode ends, and the next trial starts again from the starting point.

2. Continuing Tasks:

Definition: Continuing tasks do not have a natural endpoint; instead, the decision-making process continues indefinitely.
Structure: There is no terminal state. The agent interacts with the environment for an infinite number of time steps in a sequence.
Reward Structure: Rewards can be received at every time step, and the focus is on maximizing the average reward over an infinite horizon or minimizing the long-term cost.
Example: A self-driving car navigating on a road continuously, where the reward could be based on various criteria like safety, efficiency, and passenger comfort throughout its never-ending operation.

Summary of Key Differences:

Episode vs. Continuity: Episodic tasks have defined episodes with terminal states, while continuing tasks operate indefinitely without a terminal state.
Reward Approach: In episodic tasks, the total reward is accumulated over discrete episodes. In continuing tasks, the focus is on ongoing rewards over time, often averaged or discounted.
Application Use Cases: Episodic tasks are often simpler and suitable for problems with a clear start and finish, while continuing tasks are used for more complex environments that require ongoing interaction.

Understanding these distinctions is crucial for selecting the appropriate algorithms and approaches when dealing with different types of reinforcement learning problems.

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User

write policy, value_func, reward for both

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Assistant gpt-4o-mini

Certainly! In the context of Markov Decision Processes (MDPs), let's define what a policy, value function, and reward function looks like for both episodic and continuing tasks.

1. Episodic Tasks

Policy

Definition: A policy (\pi) in an episodic task is a mapping from states to a probability distribution over actions. It dictates the agent's behavior.
Example: [\pi(s) = P(A=a | S=s)] For instance, in a maze navigation task, (\pi) could specify the probability of taking steps Up, Down, Left, or Right when in state (s).

Value Function

Definition: The value function (V(s)) represents the expected return (cumulative rewards) received from state (s) following policy (\pi) until the episode ends.
Example: [V(s) = \mathbb{E} \left[\sum_{t=0}^{T} r_t \bigg| S_0 = s, \pi \right]] Where (T) is the time step at which the episode terminates. For instance, if the agent receives rewards of 10, 15, and -5 over an episode, then (V(s)) would be the expected return based on those rewards.

Reward Function

Definition: The reward function (R(s, a, s')) specifies the immediate reward received when transitioning from state (s) to state (s') by taking action (a).
Example: [R(s, a, s') = \begin{cases} +100 & \text{if } s' \text{ is the exit state} \ -1 & \text{if } s' \text{ is a wall} \ 0 & \text{otherwise} \end{cases}] This function gives high rewards for reaching the goal and penalties for hitting walls.

2. Continuing Tasks

Policy

Definition: Similar to an episodic task, the policy (\pi) in a continuing task maps states to a distribution over actions.
Example: [\pi(s) = P(A=a | S=s)] For a self-driving car, the policy might represent the probability of making decisions for steering, accelerating, or braking at state (s).

Value Function

Definition: The value function (V(s)) in a continuing task represents the expected return starting from state (s) and continuing indefinitely.
Example: [V(s) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r_t \bigg| S_0 = s, \pi \right]] Where (\gamma) is a discount factor (0 < (\gamma) < 1) that prioritizes immediate rewards over future rewards. This means that future rewards have less influence on the overall value due to discounting.

Reward Function

Definition: Similar to episodic tasks, but rewards can be received at every time step and can be defined for every state-action pair.
Example: [R(s, a) = \begin{cases} +1 & \text{if the action leads to fuel efficiency} \ -5 & \text{for aggressive driving behavior} \ 0 & \text{otherwise} \end{cases}] Here, rewards depend only on the current state and action rather than transitioning to a new state.

Summary

Both episodic and continuing tasks can have similar structures for policies, value functions, and reward functions.

The key difference lies in the nature of the episodes (finite in episodic vs. infinite in continuing) and how rewards are aggregated over time.

In episodic tasks, the focus is on maximizing total rewards within episodes, while in continuing tasks, it's often about maximizing the long-term average or discounted reward.